IGCSE 0607 Practice Questions — E1.12

Percentages

Total Questions: 21 — **Total Marks:** 139 Marks

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1 A shopkeeper buys a television for \$320. He sells it for \$400.
Calculate his percentage profit.

..... % **[2]**

2 **(a)** A shop has a sale.
All items are sold with a 15% discount.
Calculate the sale price of a coat originally costing \$80.

\$ ..... **[2]**

(b) A car was bought for \$4000 and sold for \$3200.
Work out the percentage loss.

..... % **[2]**

3 **(a)** Write 24 as a percentage of 60.

..... % **[2]**

(b) A coat has a normal price of \$80. In a sale, the price is reduced by 15%.
Work out the sale price of the coat.

\$ ..... **[2]**

4 **(a)** Find 15% of \$320.

\$ ..... **[2]**

(b) In a sale, the original price of a shirt is reduced by 20%.
The original price of the shirt is \$45.
Work out the sale price.

\$ ..... **[3]**

5 A shop sells a jacket for \$120. During a sale, the shop offers a 15% discount on the price of the jacket.

(a) Calculate the sale price of the jacket.

\$ ..... **[2]**

(b) The original price of \$120 gives the shop a profit of 20% on the cost price.
Calculate the cost price of the jacket.

\$ ..... **[3]**

6 **(a)** Work out 15% of \$240.

\$ ..... **[2]**

(b) A shirt costs \$40. In a sale, the price is reduced to \$32.
Calculate the percentage reduction in the price.

..... % **[2]**

(c) The price of a pair of shoes is increased by 20% to \$60.
Calculate the original price of the shoes.

\$ ..... **[2]**

7 **(a)** A shop reduces its prices by 20% in a sale.
The original price of a book is \$15.
Calculate the sale price of the book.

\$ ..... **[2]**

(b) The sale price of a jacket in the same shop is \$56.
Calculate the original price of the jacket.

\$ ..... **[3]**

(c) A different shop buys a watch for \$30 and sells it for \$42.
Calculate the percentage profit.

..... % **[2]**

8 **(a)** Alice invests \$4500 at a rate of 3.2% per year compound interest.
Calculate the total value of her investment at the end of 5 years.

\$ ..... **[3]**

(b) A shopkeeper buys a television for \$320. He sells it for \$408.
Calculate his percentage profit.

..... % **[2]**

9 **(a)** In 2021, the population of a town was 45 000.
In 2022, the population increased by 4.2%.
Calculate the population in 2022.

..... **[2]**

(b) In a sale, the price of a television is reduced by 24%.
The sale price is $342.
Calculate the original price of the television.

$ ..... **[3]**

10 Amira invests \$4500 in a savings account that pays compound interest at a rate of 2.4% per year.

(a) Calculate the total amount in Amira's account at the end of 5 years.

\$ ..... **[3]**

(b) Ben invests \$4500 in a different savings account that pays simple interest at a rate of r% per year. At the end of 5 years, Ben has the same total amount in his account as Amira has in hers.

Calculate the value of r.

$r =$ ..... **[3]**

11 **(a)** Anja invests \$8000 in an account paying compound interest at a rate of 3.5% per year.

Calculate the total value of her investment at the end of 4 years.
Give your answer correct to the nearest cent.

\$ ..... **[3]**

(b) The population of a town decreases by 4% each year.
In 2026, the population is 45 000.
Calculate the population of the town in 2023.
Give your answer correct to the nearest integer.

..... **[3]**

12 **(a)** Amira invests \$4500 in a savings account paying compound interest at a rate of 2.4% per year.
Calculate the total value of her investment at the end of 5 years.

\$ ..... **[3]**

(b) In a sale, the price of a computer is reduced by 18%.
The sale price is \$557.60.
Calculate the original price of the computer.

\$ ..... **[3]**

13 **(a)** An investment of \$8000 earns compound interest at a rate of 3.5% per year. Calculate the value of the investment at the end of 4 years.
Give your answer correct to the nearest cent.

\$ ..... **[3]**

(b) A car is bought for \$24000. Its value decreases by 12% each year. Calculate the number of complete years it takes for the value of the car to fall below \$10000.

..... years **[3]**

(c) In 2022, the price of a house was \$150000. This was a 25% increase on its price in 2017. Calculate the price of the house in 2017.

\$ ..... **[3]**

14 **(a)** Amira invests \$4000 in a bank account paying compound interest at a rate of 2.5% per year.
Calculate the total interest earned by Amira at the end of 5 years.

\$ ..... **[3]**

(b) Boris invests \$4000 in a different bank account paying simple interest at a rate of 3.2% per year.
Calculate the number of years it takes for the total value of Boris's investment to become \$4896.

..... years **[3]**

(c) Diana invests a sum of money in a bank account paying compound interest at a rate of 1.8% per year.
At the end of 6 years, her investment is worth \$6676.70.
Calculate the amount Diana originally invested, giving your answer correct to the nearest dollar.

\$ ..... **[3]**

15 **(a)** The population of a city increases by 4.5% each year.
The population is now 1.25×10^6.
Calculate the expected population in 10 years' time.
Give your answer in standard form, correct to 3 significant figures.

..... **[3]**

(b) In a sale, a washing machine is sold for \$299.25. This is a 14% reduction on the original price.
Calculate the original price.

\$ ..... **[2]**

(c) A car depreciates in value by 15% in the first year and by 10% in each subsequent year.
The value of the car after 3 years is \$17212.50.
Calculate the original value of the car.

\$ ..... **[2]**

16 The population of a city increases by $x\%$ each year.
In 2015, the population was 1 250 000.
In 2018, the population was 1 367 631.

(a) Show that the value of x is approximately 3.

[3]

(b) Assume the population continues to increase by exactly 3% each year.
Calculate the year in which the population will first exceed double its 2015 value.

..... **[4]**

17 **(a)** A shopkeeper buys a television for $\$p$. He sells it for \$960, making a percentage profit of 20% on his cost price.
Calculate the value of p.

$p = \$ \ldots\ldots\ldots\ldots\ldots$ **[2]**

(b) The shopkeeper buys another television for $\$q$. He wants to make a 15% profit on his cost price after allowing a 10% discount on the marked price.

Write down an expression, in its simplest form, for the marked price in terms of q.

$\$ \ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots$ **[3]**

(c) An investment of \$500 earns compound interest at a rate of r% per year. At the end of 10 years, the value of the investment is \$740.12.
Work out the value of r.

$r = \ldots\ldots\ldots\ldots\ldots$ **[3]**

18 **(a)** Ali invests \$5000 at a rate of 2.5% per year compound interest. Calculate the value of his investment at the end of 3 years.
Give your answer correct to the nearest cent.

\$ ..... **[2]**

(b) Beatrice invests a sum of money at a rate of r% per year compound interest.
After 5 years, the value of her investment has increased by 30%.
Calculate the value of r.

$r =$ ..... **[3]**

(c) The value of a car depreciates by 12% each year.
After 3 years, the value of the car is \$17 036.80.
Calculate the original value of the car.

\$ ..... **[3]**

19 **(a)** A car's value depreciates by 15% in the first year.
In each subsequent year, its value depreciates by 8%.
After exactly 3 years, the value of the car is \$17 986.
Calculate the original value of the car.

\$ ..... **[4]**

(b) Calculate the overall percentage decrease in the car's value over the 3 years.
Give your answer correct to 1 decimal place.

..... % **[2]**

(c) The population of a city is P.
The population increases by r% each year.
After 12 years, the population has exactly doubled.
Calculate the value of r.

$r =$ ..... **[3]**

20 **(a)** A car depreciates in value by 15% each year.
After 3 years, the value of the car is \$12 282.50.
Calculate the original value of the car.

\$ ..... **[3]**

(b) An antique clock was bought for \$15 000. Its value increases exponentially at a rate of x% per year. After 5 years, its value is \$21 500.
Calculate the value of x, giving your answer correct to 1 decimal place.

$x =$ ..... **[3]**

(c) A house is bought for \$250 000. Its value increases by 4% in the first year, decreases by 2% in the second year, and increases by 5% in the third year. Calculate the overall percentage increase in the value of the house over the 3 years.
Give your answer correct to 2 decimal places.

..... % **[4]**

21 **(a)** A population of bacteria increases by $r\%$ every hour.
The population doubles in 6 hours.
Calculate the value of r, giving your answer correct to 3 significant figures.

$r =$ ..... **[3]**

(b) A merchant buys goods for $\$X$. He sets the selling price such that he makes a 40% profit on the cost price. During a sale, he gives a 15% discount on this selling price. The final sale price is \$142.80.
Calculate the value of X.

$X =$ ..... **[3]**

(c) An amount of $\$P$ is invested at a rate of $x\%$ per year simple interest. The total simple interest earned in 5 years is \$250.
The same amount $\$P$ is invested at a rate of $x\%$ per year compound interest. The total compound interest earned in 2 years is \$102.50.
Find the value of P and the value of x.

$P =$ \$ ..... , x = ... **[5]**